

Homework 7, due Oct 22 before/in class

1. Estimate deflection angle for an electron of energy $\epsilon = 1$ keV passing at a distance l one micrometer from the proton.
2. It's a bright sunny day, Sun in the zenith. You are holding a parasol of radius $r = 50$ cm. Find force from the Sun (assume complete absorption). Look up luminosity of the Sun and distance to the Sun.
3. EM wave is propagating along z and is polarized along x , so that its vector potential is

$$\mathbf{A} = A_0 \{\cos(\omega t - k_z z), 0, 0\}, \omega = k_z c \quad (1)$$

Calculate averaged (over period of oscillation) Poynting flux.

Represent this linearly polarized wave as a sum of two circularly polarized waves, write vector potential for $\mathbf{A}_R$ and $\mathbf{A}_L$, the right and left-polarized components

There are media where Left and Right circular polarizations propagate with different speeds (called gyrotropic). After passing through such a media one of the waves (does not matter which, right or left) is delayed in phase by π (so that it becomes $\propto \omega t - k_z z + \pi$). Find new polarization after the wave left that gyrotropic medium.

4. EM wave is propagating along z and is polarized along x , so that its vector potential is

$$\mathbf{A}_1 = A_0 \{\cos(\omega t - k_z z), 0, 0\}, \omega = k_z c \quad (2)$$

Another EM wave propagating along z and is polarized along x , is added

$$\mathbf{A}_2 = A_0 \{\cos(\omega t - k_z z), 0, 0\}, \omega = k_z c \quad (3)$$

(the same). Calculate averaged (over period of oscillation) Poynting flux of two waves together.